

Multi-Modal Model Training Metrics Plan

Complete Metric Table for 4 Classification Types

Metric	Text	Image	Audio	Video
Primary Modality	Tweets, comments, posts	Photos, drawings, scans	Speech, commands, audio clips	Multi-frames, video segments, transcripts
Model Architecture	DistilBERT/RoBERTa	CNN (ResNet, EfficientNet)	Wav2Vec, CNN on spectrograms	CNN-LSTM, ViT, SlowFast
Typical Input Size	512-1024 tokens	224×224 pixels	16kHz audio, 1-30 seconds	16-64 frames, 100ms-2fps
Batch Size (GPU 8GB)	32-64	16-32	16-32	8-16
VRAM Required	4-6GB	6-8GB	4-6GB	8-12GB
Training Time (500 samples)	3-5 min	5-8 min	4-7 min	8-12 min
Training Time (16K samples)	45-90 min	2-3 hours	2-3 hours	4-6 hours
Dataset Size (Minimum)	500 samples	1,000 images	500 audio clips	200 video segments
Dataset Size (Recommended)	5,000-10,000	10,000-20,000	2,000-5,000	1,000-3,000 video segments

Metric	Text	Image	Audio	Video
Dataset Size (Optimal)	50,000+	50,000+	10,000+	5,000+ with diversity
Epochs Needed	1-3	5-10	10-20	10-15
Typical Accuracy	85-95%	75-90%	70-85%	75-88%
Complexity	Low-Medium	Medium-High	Medium	High
Preprocessing	Tokenization, lowercasing	Resize, normalize, augment	MFCC, spectrogram, augmentation	Frame extraction, temporal augment, audio extraction

Dataset Size Recommendations by Modality

Text Classification (Cyberbullying)

- **Minimum:** 500 samples (quick demo achievable)
- **Recommended:** 5,000-10,000 samples
- **Optimal:** 50,000+ samples
- **Note:** 16,851-47,692 tweets available

Image Classification

- **Minimum:** 1,000 images (per class if multi-class)
- **Recommended:** 10,000-20,000 images
- **Optimal:** 50,000+ images (ImageNet-scale)
- **Note:** Requires data augmentation for best results

Audio Classification

- **Minimum:** 500 audio clips (1-30 seconds each)
- **Recommended:** 2,000-5,000 audio clips

- **Optimal:** 10,000+ audio clips
- **Note:** Must balance across classes (bullying vs clean speech)

Video Classification

- **Minimum:** 200 video segments (1-5 seconds each)
- **Recommended:** 1,000-3,000 video segments
- **Optimal:** 5,000+ video segments with diversity
- **Note:** Requires frame extraction and temporal feature engineering

Key Planning Insights

1. Text is Easiest & Fastest

- Project already has text datasets working
- Quickest training time (3-5 min for 500 samples)
- Best accuracy results (85-95%)

2. Video Requires Most Resources

- Highest VRAM demand (8-12GB)
- Longest training time
- Needs most dataset diversity
- **Consider:** Extract frames only, not full video processing

3. Audio Needs Moderate Resources

- Similar VRAM to text
- Needs audio feature extraction (MFCC, spectrograms)
- **Consider:** Use existing speech datasets, not raw audio

4. Image Needs Medium Resources

- CNNs well-established for image tasks
- Requires data augmentation (rotation, flip, color jitter)
- **Consider:** Use pretrained ImageNet models (ResNet50, EfficientNet)

Resource Planning Checklist

Before Starting Multi-Modal Training:

- ☐ **VRAM:** 24-32GB total (or train sequentially)
- ☐ **Disk Space:** 10GB+ for all datasets
- ☐ **Time:** 8-15 hours (sequential) or 2-3 hours (parallel)
- **Datasets:**
 - ☐ Text: 5,000-50,000 samples (already have)
 - ☐ Images: 10,000-50,000 images
 - ☐ Audio: 2,000-10,000 audio clips
 - ☐ Video: 1,000-5,000 video segments
- **Hardware:** GPU with 24GB+ VRAM OR train modalities one-at-a-time

Recommendation for this Project

Given existing text datasets (16K-47K samples) and successful quick demo:

Phase 1: Text Only (Already Complete)

- Used existing 16K-47K Twitter/Toxicity datasets
- Achieve 85-95% accuracy
- Training in 15-90 minutes

Phase 2: Add Image Modality

- Collect 10,000+ labeled images
- Use pretrained ResNet50/EfficientNet
- Add 2-3 hours training time
- Accuracy: 75-90%

Phase 3: Add Audio Modality

- Collect 2,000-5,000 audio clips
- Extract MFCC/spectrogram features

- Add 2-3 hours training time
- Accuracy: 70-85%

Phase 4: Add Video Modality

- Collect 1,000-3,000 video segments
- Extract frames + temporal features
- Add 4-6 hours training time
- Accuracy: 75-88%

Final Combined Model

- **Total Training Time:** ~15-25 hours (sequential)
- **Total Datasets:** ~66K+ samples across 4 modalities
- **Expected Accuracy:** 75-95% (weighted by modality)

Strategic Recommendation

1. **Phase 1:** Optimize text model with full 16K-47K datasets
2. **Phase 2:** Add Image modality if visual cyberbullying detection needed (memes, posted images)
3. **Phase 3:** Add Audio modality if detecting audio-based bullying (harassing calls, voice messages)
4. **Phase 4:** Add Video modality only if absolutely needed (rare for cyberbullying)